

Predictive Customer Churn Analysis Report

EXECUTIVE INTELLIGENCE BRIEF • PORTFOLIO COHORT: 250,000 ACCOUNTS

250,000

TOTAL CUSTOMER COHORT

36.98%

AVERAGE CHURN PROBABILITY

1. Executive Summary

This diagnostic report provides a data-driven evaluation of retention risks across a global customer portfolio of 250,000 accounts. Advanced predictive modeling indicates an alarming baseline churn probability of **36.98%**. Left unchecked, this level of volatility poses substantial threats to stable recurring revenues. The primary goal of this analysis is to identify operational bottlenecks, customer touchpoint friction thresholds, and infrastructure vulnerabilities to inform rapid tactical interventions.

2. Key Behavioral Triggers & Threshold Analysis

The Operational 'Friction Point': Customer Support Escalations

Analysis of support call histories yields a stark, non-linear predictive threshold. Up to 3 support calls, customer churn risk remains perfectly stable and suppressed at approximately **32%**. However, there is a catastrophic failure point between the 3rd and 4th interaction:

- **0–3 Support Calls:** Risk is managed and flat (~32% probability).
- **4+ Support Calls:** Risk instantly surges to **~70% probability**, establishing a permanent high-risk plateau that persists all the way through to 10 calls.

Critical Insight: The 4th customer support interaction represents an absolute breaking point in the customer lifecycle. This strongly implies that issues are either remaining unresolved across multiple touches, or the cumulative operational friction triggers an active decision to switch providers.

Pricing & Rate Sensitivity

High-value accounts exhibit steep attrition markers. A dense concentration of risk sits squarely within the \$100 to \$150 monthly charge spectrum, where accounts regularly cross into the 85%–100% churn probability band. This signals that premium tier clients expect flawless service delivery, and are highly sensitive to billing friction.

3. Granular Segmentation & Risk Decomposition

By isolating customer attributes through dimensional decomposition, we have pinpointed exactly where structural churn is heavily concentrated.

Risk Hierarchy Level	Segment Demographics	Volumetric Churn Impact	Relative Vulnerability
1. Primary Region	North America	26,611	Highest Global Volume
2. Core Infrastructure	North America → Fiber Optic	12,014	45.1% of Regional Attrition
3. Contract Framework	North America → Fiber Optic → Month-to-Month	11,397	94.8% of Infrastructure Risk

Conversely, users on structured **One-Year Contracts** within the exact same regional and infrastructure footprint represent a mere fraction of the risk (**392 accounts**). This validates that contract structure, rather than infrastructure alone, serves as a primary driver of retention security.

4. Network Infrastructure and Data Utilization

Data volume distribution confirms that higher bandwidth services are coupled with severe retention exposure.

- **Fiber Optic:** Leads overall network consumption at over **13 Million GB**, while simultaneously holding the highest churn concentration.
- **DSL Infrastructure:** Accounts for approximately **9 Million GB** with moderate stability.
- **5G Networks:** Remains the lowest consumer tier at roughly **6 Million GB**.

5. Recommended Tactical Actions

1. **Automated Support Desk Early Warning System (EWS):** Configure real-time CRM flags on customer records as soon as a 3rd support call is logged. The 4th incoming interaction must bypass standard queues and be automatically routed to a senior retention specialist.
2. **Targeted Contract Conversion Campaigns:** Deploy highly structured loyalty incentives specifically targeting North American Fiber Optic users currently on Month-to-Month agreements. Transitioning these accounts to structured 1-Year plans reduces risk by over 90% based on cohort metrics.
3. **Premium Plan Billing & Value Audits:** Audit service experiences for any account billing over \$100/month. Introduce personalized value-add touches to justify the premium price point and clear the dense 85%+ churn risk cluster.